

Bug World

Task Sheet 4 (update of 3)

Software Engineering Project

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Overview. This task adds to the original semester project task described in the [Bug World Task Sheet](#). Functionality described below adds to that original task, it does not remove any.

Task. Write a bug program which wins in a tournament. Some sample .bug and .buggy files are [available on the course site](#).

Extras: The simulator behavior is changed over the original one in two respects:

- Day/nighttime: Every 1,000 ticks, daytime changes to nighttime, and vice versa. Simulation starts with daytime. Daytime behavior of the world is as before, during nighttime, thought, markers are unreadable (i.e., any sensing will not report a marker even if there is one)
- Bugs understand an additional command:

kill *pos*

where *pos* is a standard relative position as in other position-sensitive commands.

Interpretation: If there is a foe bug at the position indicated, that bug will be removed from the world, and any food carried by that bug will be dropped at that position. If there is no foe at the position scanned then the scanning bug will lose its food carried, if any.

Documentation update: "Surrounding Rule".

- The simulator automatically kills a bug when it is surrounded by enemy bugs. This rule is always active and is independent of the `kill` opcode.
- **Trigger condition:** A bug is killed if **5 or more of its 6 adjacent cells** are occupied by enemy bugs (bugs of the opposite color).
- **On death:**
 - The dead bug is removed from the world.
 - **3 food units** plus any food the bug was carrying are dropped on its cell.
- **When it is checked:** After every `move`, the simulator checks the moved bug and all 6 of its neighbours for the surrounding condition.

- **Implication for strategy:** A single bug surrounded on 5 sides by enemies dies instantly, regardless of what instruction it is executing. Clustering your own bugs defensively or driving enemies into surrounded positions are both valid strategic considerations.

Evaluation: TAs will run the bugs received in a tournament where each bug code submitted will be run against 3 separate bug codes (i.e., not submitted by students). Each round consists of two rounds: once with the bug under test as red, and once as black. A win yields 2 points, a draw 1 point, a loss 0 points. All points will be summed up. Bugs get ranked according to their achievement, and grade points are assigned to students in descending order, starting with 40, so: a ranking position N yields 40-N grade points. Exception: Same positions yield same points.

Submission:

- A single valid .bug or .buggy file
- Per person (not as team)
- By mail to TA Pico, Mateo mpico@constructor.university
- Deadline: ~~2026-04-28~~ 23:59 CEST 2026-05-07 23:59 CEST